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Hello,

The following submission has been edited.

Track Name: IJCACI2026

Paper ID: 751

Paper Title: The Influence of Visualization Design on Decision Making

Abstract:

The transformation of raw numerical data into clear and meaningful visual formats has significantly reshaped decision-making across modern organizations. In its unprocessed tabular form, data is often difficult to interpret, especially under conditions of time pressure, uncertainty, and high cognitive demand. Data visualization addresses this challenge by converting complex numerical relationships into structured, intuitive, and visually guided representations that align with human cognitive processes. This research examines how four key dimensions of visualization design—interactivity, color hierarchy, narrative structure, and visual complexity—affect decision-making quality, speed, and accuracy. The study combines empirical insights from ten global enterprises across diverse industries with a systematic review of twenty-one academic studies spanning cognitive psychology, information science, human-computer interaction, and management.

A mixed-methods approach was employed, including descriptive statistics, Chi-Square hypothesis testing at a 0.05 significance level, and comparative performance benchmarking. Results consistently indicate that well-designed and interactive visualizations enhance decision-making efficiency by reducing cognitive effort and improving clarity. Organizations using advanced visualization practices experienced up to a 67% increase in decision accuracy and a reduction of up to 74 days in decision cycles. In contrast, those relying on static or overly complex visuals saw only modest improvements.

All statistical tests yielded strong significance ($p < 0.01$), reinforcing the study's findings. The research also highlights applications across sectors such as finance, healthcare, retail, governance, and energy, while pointing toward future trends including AI-driven visualization, real-time data representation, and ethical considerations.

Visualization design drives decision intelligence and performance, not just visual appeal in organizations.

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Submission Questions Response:

1. Conflict of interest
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6. Authors Contributions

Saumya Jain - study conception and design, data collection, analysis

Shikha Tuteja - Conceptualization, Supervision, Methodology, Writing – Review & Editing,

7. Conference Rules and Guidelines

Agreement accepted

Thanks,
CMT team.

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